Alexandre Fournier Montgieux

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Experience

PhD in Machine Learning & Computer Vision — Palaiseau, France
CEA LIST – Nano Innov — *Nov. 2023 – Nov. 2026 (Ongoing)*

- Research Focus: Generation of Realistic and Diversified Synthetic Training Datasets for Face Recognition.
- Released: **Face Recognition**, bias auditing framework, synthetic benchmark and training sets; Image Contrastive **Generative Approach**; Frugal **Facial Demographic Annotator** with diversified training and evaluation sets; **Recommendation Systems** unified auditing framework; 3D annotation approach for **Autonomous Driving** .

Referent Teacher – AI Class, Master BDMA — Gif-sur-Yvette, France
Centrale Supélec — *2024 – 2026*

- Designed and delivered course sessions on deep learning and computer vision for M2 students.

Intern: Image Generation through GAN — Palaiseau, France
CEA LIST – Nano Innov — *Apr. 2023 – Sep. 2023*

- Research Focus: Generation of Diversified Synthetic Training Datasets with GAN approaches.
- Delivered a **Diversified Synthetic Training Set** for Face-Recognition models

Intern: Image Processing for automated lithology characterization — Clamart, France
Schlumberger / SLB — *Feb. 2022 – Aug. 2022*

- Research Focus: Segmentation and regression approaches for grain size estimation in rocks from microscopic 2D pictures.
- Delivered: **Image Processing Segmentation** approaches and **Frugal Deep Classification** Models.

Front-End Developer: BackOffice Team — Paris, France
Stockly — *Aug. 2021 – Fev. 2022*

- Participated in the creation and development of the **deployed BackOffice** platform (coded in Rust and Angular). The platform aimed to improve the company's logistics management, bringing order and supplier workflows into a single working environment alongside existing systems.

Publications

Diff-CA: Separating Common and Salient Factors with Diffusion Models — Neurips 2026 (Spotlight)

- Soumm M., **Fournier-Montgieux, A.**, He Y., Gori P., & Newson A.

MVAT: Multi-View Aware Teacher for Weakly Supervised 3D Object Detection — WACV 2026

- Lahlali, S., **Fournier Montgieux, A.**, Granger, N., Le Borgne, H., & Pham, Q. C.

Quantifying User Coherence: A Unified Framework for Cross-Domain Recommendation Analysis WWW 2026

- **Fournier-Montgieux, A.**,Soumm, M., Popescu, A., & Delezoide, B.

Reliable and Reproducible Demographic Inference for Fairness in Face Analysis — arXiv 2025

- **Fournier-Montgieux, A.**, Le Borgne, H., Popescu, A., & Luvison, B.

Fairer Analysis and Demographically Balanced Face Generation for Fairer Face Verification WACV 2025

- **Fournier-Montgieux, A.**, Soumm, M., Popescu, A., Luvison, B., & Le Borgne, H.

Research Projects

SHARP Project | Inter-lab Collaboration — 2023 – 2026
Research Visit | CERTH, Thessaloniki, Greece — April – May 2024

Education

MSc. Mathematics, Vision, Learning (MVA) — Cachan, France
ENS Paris-Saclay – Mention:Très Bien (With Honors) — *2022 – 2023*

Engineering Degree in Digital Sciences — Paris, France
Télécom Paris – GPA:4/4 — *2019 – 2023*

Preparatory Classes (MPSI / MP* / MP) — Paris, France
Lycée Janson de Sailly — *2017 – 2019*